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Genome-wide analysis and identification of *HAK* potassium transporter gene family in maize (*Zea mays* L.)

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Abstract The high-affinity K^+ (HAK) transporter gene family constitutes the largest family that functions as potassium transporter in plant and is important for various cellular processes of plant life. In spite of their physiological importance, systematic analyses of ZmHAK genes have not yet been investigated. In this paper, we indicated the isolation and characterization of ZmHAK genes in whole-genome wide by using bioinformatics methods. A total of 27 members (ZmHAK1–ZmHAK27) of this family were identified in maize genome. ZmHAK genes were distributed in all the maize 10 chromosomes. These genes expanded in the maize genome partly due to tandem and segmental duplication events. Multiple alignment and motif display results revealed major maize ZmHAK proteins share all the three conserved domains. Phylogenetic analysis indicated *ZmHAK* family can be divided into six subfamilies. Putative *cis*-elements involved in Ca^{2+} response, abiotic stress adaption, light and circadian rhythms regulation and seed development were observed in the promoters of *ZmHAK* genes. Expression data mining suggested maize *ZmHAK* genes have temporal and spatial expression pattern. In all, these results will provide molecular insights into the potassium transporter research in maize.

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Abbreviations

GA_3	Gibberellin
HAK	High-affinity K^+
NAA	Naphthaleneacetic acid
ORF	Open reading frame
KCO	K^+ channel openers
KT	Cytokinin

Introduction

Potassium (K^+) is an essential nutrient for plant in various cellular processes, which is involved in enzyme function, stomata movements, cell elongation, signal transduction, osmoregulation, and maintenance of the plasma membrane potential [1–8]. There were two transport systems for K^+ uptake in plants, which referred to as potassium channels and transporters [9, 10]. In plants, potassium transporters were divided into four multi-gene families: the KT (K^+ transporter)/HAK (high-affinity K^+)/KUP(K^+ up-take) family, Trk/HKT family, KEA (Kt/Ht antiprotons) family, and CHX (cation/hydrogen exchanger) family [11, 12]. Among them, KT/HAK/KUP family, which was also called KT, KUP or HAK transporter family, was the largest family [13].

The members of KT/HAK/KUP family were firstly found in Fungi and bacteria [14]. Based on sequence homology analysis with *KUP* from bacteria and *HAKs* from fungi, *KT/HAK/KUP* genes had been cloned from pepper (*Capsicum annum*), lotus (*Lotus japonicus*), seagrass (*Cymodocea nodosa*), grapevine (*Vitis vinifera*), tomato (*Lycopersicon esculentum*), rice (*Oryza sativa*) and *Arabidopsis thaliana* [4, 11, 12, 15–23]. HAK transporters from these plants played important role various aspects of plant life [24]. Some members of this family had function in high affinity K^+ uptake and were up-regulated in response to K^+ starvation, such as *OsHAK1* in rice [24], *AtHAK5* in *A. thaliana* [13, 25], *HvHAK1* in barley [4] and *LeHAK5* in tomato [19]. On the other hand, some members of these transporters, such as *OsHAK7* and *OsHAK10* had a role in low-affinity K^+ transport supplementing potassium channels [24].

Previous studies had shown that HAK transporters were expressed throughout the plant, including in flower, leaf, and stem [18, 24, 26, 27]. These results suggested that this family did not simply mediated primary K^+ uptake from soil. Some transporters might also involve in regulatory processes. For example, an *Arabidopsis atkup2* mutation was shown to cause short hypocotyl phenotype [28], while knock-out *TRH1/AtKT3* shown tiny root hair phenotype in *Arabidopsis* [29]. And *AtKT3* was also involved in auxin transport [30]. Some transporters might also be involved in salinity response, such as *AtHAK1*, *AtHAK6*, *AtHAK2 in Arabidopsis* [31], *HvHAK1* in barley [4], *PhaHAK1* in reed plants [32]. Recent study indicated that some rice *HAK* genes involved in hormone response, such as *OsHAK7* involved in naphthaleneacetic acid (NAA) response; *OsHAK8* involved in NAA, KT response; *OsHAK16* involved in gibberellin (GA_3), KT response; *OsHAK17*, *OsHAK27* involved in NAA, GA_3 and KT response [12].

Results of B73 genome sequencing were released and published in Science magazine on November 20, 2009 [33]. Based on the sequence resources of maize genome, we investigated maize *HAK* genes in whole genome scale. In this paper, we aimed at analyzing maize *HAK* genes number, their distribution on genome, expansion pattern, characteristic of motif distribution, evolutionary relationship, *cis*-elements in promoter regions and expression profiles.

Materials and methods

Isolation of *HAK* genes in maize

The sequences of 27 rice [34] and 13 *Arabidopsis* [8, 11] HAK transporter proteins were downloaded from the TIGR database (http://rice.plantbiology.msu.edu) and TAIR database (http://www.arabidopsis.org). To obtain all the *HAK* genes in maize, BLASTP searches were performed in the Maize sequencing database (http://www.maizesequence.org/index.html), the maizeGDB database (http://www.maizegdb.org/), and NCBI database (http://www.ncbi.nlm.nih.gov/) with the *Arabidopsis* and rice HAK transporter proteins as queries. First, we selected sequence as a candidate ZmHAK protein if it satisfied with $E < 10^{-10}$. And then, the tool of Pfam was used to identify the K^+ transporter domain (PF02705) of all the candidate ZmHAK proteins [35]. The full-length cDNA sequences of *ZmHAK* genes in maize were downloaded from the Maize sequencing database.

Gene structure analysis of maize *HAK* genes

The information of *ZmHAK* genes, including open reading frame (ORF) length, chromosomal location, full length cDNA sequences were obtained from the B73 maize sequencing database (http://www.maizesequence.org/index.html). Exons and introns structures of *ZmHAK* genes were determined by using GSDS (http://gsds.cbi.pku.edu.cn/) [36].

Motif display and phylogenetic analysis of HAK proteins

Multiple expectation maximization for motif elicitation (MEME) utility program [37] was used to display motifs of ZmHAK proteins. The matrix for phylogenetic analysis included the 13, 27 and 27 *HAK* genes from *Arabidopsis*, rice, and maize. The amino acid sequences of all the proteins were aligned using ClustalX2.0 [38] and the unrooted phylogenetic tree was constructed by the neighbor-joining (NJ) method using MEGA4 software [39].

Promoter regions analysis of *ZmHAK* genes

To investigate *cis*-elements in promoter sequences of *ZmHAK* genes, 2 kb of B73 genomic DNA sequences upstream of the initiation codon (ATG) were downloaded from NCBI. And then the PLACE (http://www.dna.affrc.go.jp/PLACE/) was used to analyse *cis*-elements in the promoters [40].

Analysis of *ZmHAK* genes expansion patterns

Tandem and segmental duplication have impact on gene family amplification [41]. Tandem duplications were characterized as multiple members of this family on condition that five or fewer protein-coding genes between them [41]. We considered that paralogs are segmental duplications if they reside within a region of conserved protein-coding genes [34]. In order to identify tandem and segmental duplications, a similar method with Yang's and Maher's [34, 42] was used. To classify apparent expansions of *ZmHAK* gene family, we investigated the physical locations of all members of this family in maize. Firstly, we identified the paralogous *ZmHAK* genes at the terminal nodes of the phylogenetic tree. Secondly, 20 protein-coding genes flanking paralogs each (10 protein-coding genes upstream and downstream of each pair of paralogs) were obtained from http://www.maizesequence.org/index.html.

Expression profiles investigation of maize *HAK* genes

Transcript levels of *ZmHAK* genes were analyzed by multiple methods [41]. Firstly, the coding sequences of *ZmHAK* genes were used to query the NCBI EST database (http://www.ncbi.nlm.nih.gov/dbEST/) using megablast tool. Parameters were as followings: maximum identity >95 %, length >200 bp and *E* value $<10^{-10}$. Secondly, expression data of *ZmHAK* genes were retrieved using the maize sequencing by synthesis (SBS), whole-genome shotgun (WGS) database (http://mpss.udel.edu/maize_WGS/) [43].

Results

The HAK potassium transporters family in maize

After carefully surveying the maize genome, 27 members were defined as *ZmHAK* genes (Table 1). There was no *ZmHAK* gene homologous to *OsHAK6* but two *OsHAK16* homologous genes named *ZmHAK16a* and *ZmHAK16b.* The length of ZmHAK transporters ranged from 642 aa (amino acids) to 921 aa. BLAST analysis against the Pfam database showed that all of them belonged to the ZmHAK family. The tool of TMHMM integrated in the program InterProScan [44] was used to predict trans-membrane segments for ZmHAK transporters. Each ZmHAK protein contained at least 9 trans-membrane segments.

All the predicted ZmHAK proteins had a typical "k_trans" domain (PF02705), which was specific to HAK

Table 1 List of *HAK* genes in maize

Gene Name	Chr[a]	ORF length	Protein length	Protein ID	TMS[b]	PL[c]
ZmHAK1	2	2,343	781	GRMZM2G093826_P01	12	PM
ZmHAK2	8	2,196	732	AC233953.1_FGP005	11	PM
ZmHAK3	8	2,436	812	GRMZM2G477457_P01	10	PM
ZmHAK4	4	2,094	698	GRMZM2G425999_P01	9	CTM
ZmHAK5	3	2,319	773	GRMZM2G084779_P01	11	PM
ZmHAK7	7	2,367	789	GRMZM2G020766_P01	13	PM
ZmHAK8	1	2,382	794	GRMZM2G173387_P01	12	PM
ZmHAK9	7	2,379	793	GRMZM2G166738_P01	11	PM
ZmHAK10	9	2,436	812	GRMZM2G443728_P02	10	PM
ZmHAK11	2	2,427	809	GRMZM2G005040_P01	14	PM
ZmHAK12	10	2,376	792	GRMZM2G009353_P02	13	PM
ZmHAK13	6	2,373	791	GRMZM2G146140_P02	13	PM
ZmHAK14	7	2,499	833	GRMZM2G139931_P01	12	PM
ZmHAK15	2	2,541	847	GRMZM2G121063_P02	12	PM
ZmHAK16a	1	2,511	837	GRMZM2G084486_P01	11	PM
ZmHAK16b	1	2,421	807	GRMZM2G097505_P01	11	PM
ZmHAK17	7	1,926	642	GRMZM2G088964_P01	11	ER
ZmHAK18	7	2,361	787	GRMZM2G086389_P02	14	PM
ZmHAK19	10	2,229	743	GRMZM2G021725_P01	10	PM
ZmHAK20	2	2,409	803	GRMZM2G395267_P01	11	PM
ZmHAK21	1	2,508	836	GRMZM2G438960_P01	12	PM
ZmHAK22	2	2,763	921	GRMZM2G409771_P01	10	PM
ZmHAK23	7	2,556	852	GRMZM2G036916_P01	12	PM
ZmHAK24	6	2,307	769	GRMZM2G120163_P01	11	PM
ZmHAK25	5	2,319	773	GRMZM2G375116_P01	13	PM
ZmHAK26	2	2,220	740	NP_001148930	10	PM
ZmHAK27	1	2,448	816	GRMZM2G455817_P01	11	PM

PM plasma membrane, *CTM* chloroplast thylakoid membrane, *ER* endoplasmic reticulum

[a] Chromosome number in which the gene resides

[b] TMS: number of transmembrane segments posses

[c] Localization of ZmHAK protein supported by PSORT (http://psort.nibb.ac.jp)

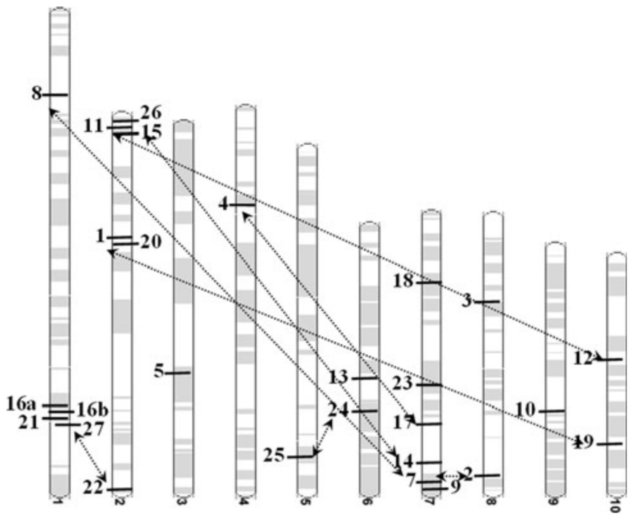

Fig. 1 Chromosomal localization of maize *HAK* genes. Segmental duplicates, including *ZmHAK1/ZmHAK19*, *ZmHAK2/ZmHAK7*, *ZmHAK4/ZmHAK17*, *ZmHAK8/ZmHAK9*, *ZmHAK11/ZmHAK12*, *ZmHAK14/ZmHAK15*, *ZmHAK22/ZmHAK27*, and *ZmHAK24/ZmHAK25* were indicated by bidirectional arrows in figure

potassium transporters family members. Twenty-five of the 27 ZmHAK proteins show localization on plasma membranes by a PSORT analysis (http://psort.nibb.ac.jp/), a location that was best suited for their function as transporter. The remaining two proteins ZmHAK4 and ZmHAK17 showed localization on chloroplast thylakoid membrane and endoplasmic reticulum (Table 1).

The 27 *ZmHAK* genes were distributed on ten maize chromosomes: chromosomes 2 and 7 each contains six genes, chromosome 1 contained five genes, chromosomes 6, 8 and 10 each contained two genes and chromosomes 3, 4, 5, and 9 each contained one genes (Fig. 1). The full-length cDNA sequences were compared with the corresponding genomic DNA sequences to determine the numbers and positions of exons and introns within each *ZmHAK* gene (Fig. 2) by using GSDS (http://gsds.cbi.pku.edu.cn/chinese.php) [36]. The coding sequences of all the *ZmHAK* genes were disrupted by introns, with numbers varying from two to nine.

Motif analysis and protein architecture

Based on the MEME web server, the motif distribution in ZmHAK proteins were analyzed. Three putative conserved motifs were identified, with 50, 50, 34 amino acids, respectively. Among these three motifs, motif 1, 2, 3 has 19, 32, 22 conserved amino acid residues, respectively.

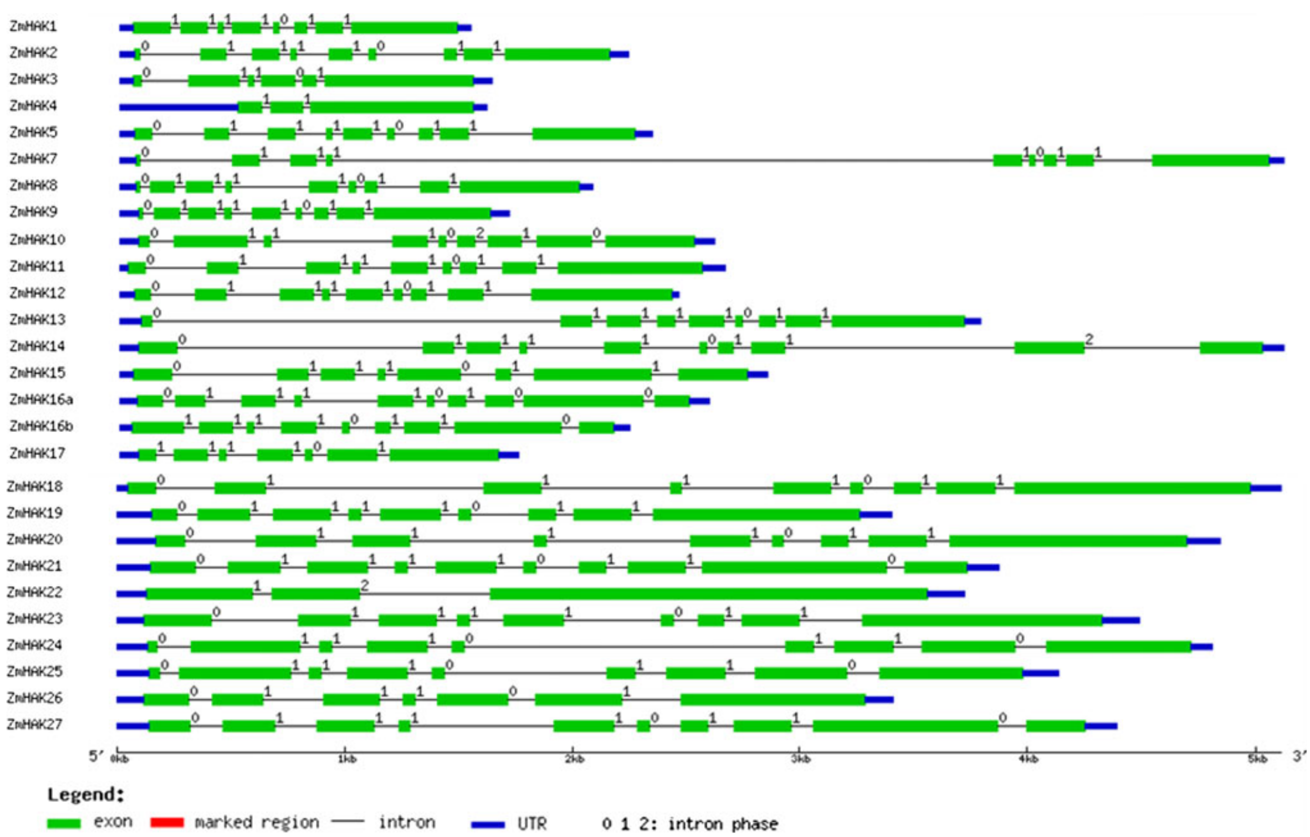

Fig. 2 Gene structure of maize HAK family. Exons and introns were showed by filled boxes and single lines, respectively. UTRs were displayed by *thick blue lines* at both ends. Introns phases 0, 1 and 2 were indicated by numbers *0*, *1*, and *2* in figure

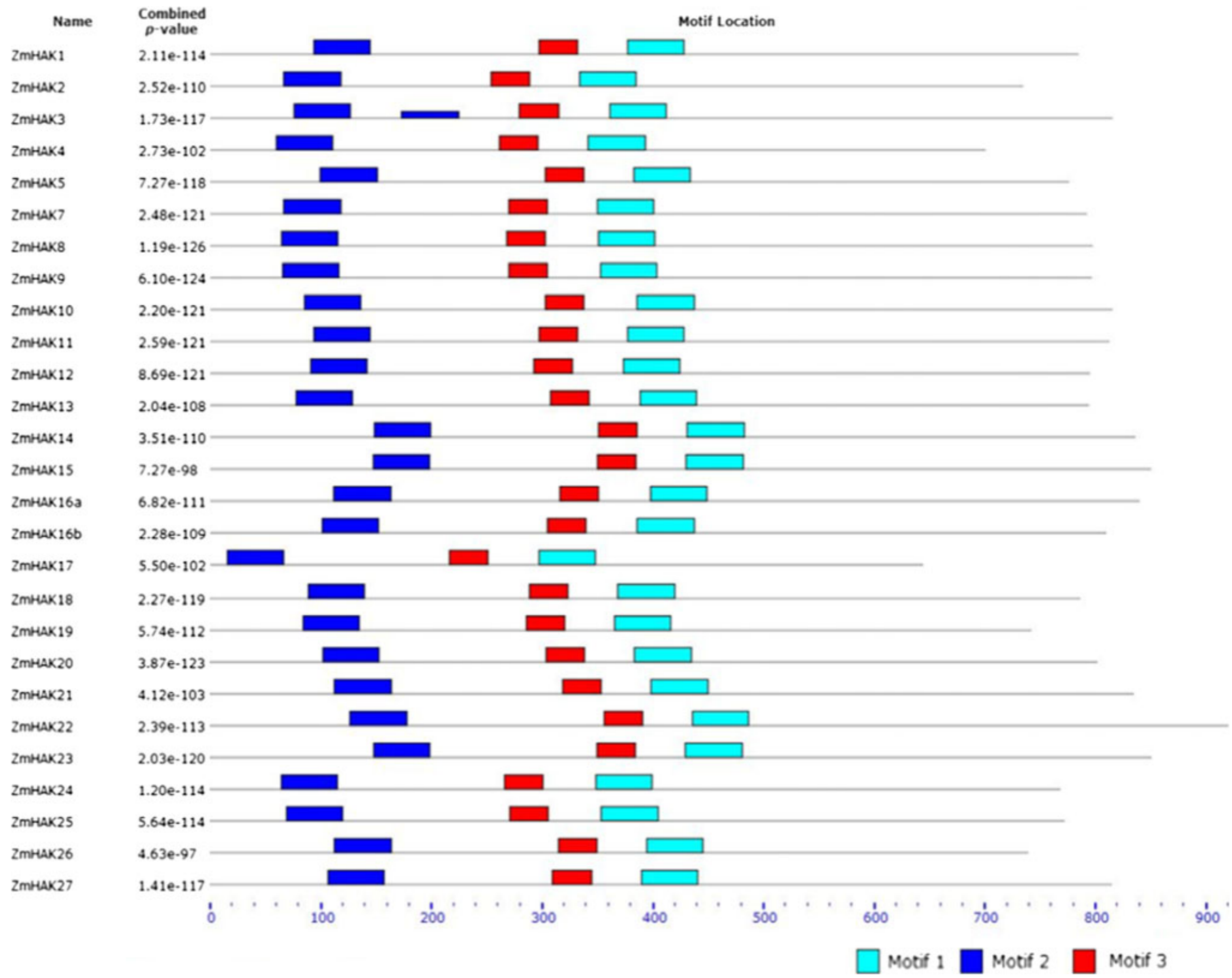

Fig. 3 Putative motif distribution in maize HAK proteins. Motifs of ZmHAK proteins were investigated by MEME web server

All these three putative conserved motifs were present in all the ZmHAK members and motif 1 appeared twice in ZmHAK3 (Fig. 3).

Phylogenetic analysis

In order to explore the anagenesis pattern and phylogenc relationships among *ZmHAKs* in maize (27 genes), rice (27 genes) and *Arabidopsis* (13 genes), the amino acid sequences of HAK genes were aligned by clutalX and a phylogenetic tree was constructed with NJ method (Fig. 4). All the proteins fell into four major clusters (I, II, III, and IV). Cluster I contained 18 members (with 9, 8, 1 members of maize, rice, *Arabidopsis*, respectively) and can be further divided into subcluster IA and IB, and a diverge member (AtHAK5). Cluster II included 24 members (with 9, 9, 6 members of maize, rice, *Arabidopsis*, respectively) and can be separated into subcluster IIA and IIB. Cluster III contained 18 members (with 6, 6, 6 members of maize, rice, *Arabidopsis*, respectively) and was composed of subclusters IIIA and IIIB. Cluster IV has 7 members (with 3, 4 members of maize, rice, respectively), which could be further divided into subclusters IVA and IVB.

Genome duplication of *ZmHAK* genes

Based on phylogenetic results (Supplementary Fig. 1), nine paralogs were identified in *ZmHAK* genes. According to the B73 maize genome annotation results, *ZmHAK16a* and *ZmHAK16b* were located in one tandem duplicate. While segmental duplications might contribute to other seven paralogous genes, *ZmHAK1/ZmHAK19*, *ZmHAK2/ZmHAK7*, *ZmHAK4/ZmHAK17*, *ZmHAK11/ZmHAK12*, *ZmHAK14/ZmHAK15*, *ZmHAK22/ZmHAK27*, and *ZmHAK24/ZmHAK25*. There were conserved protein-coding genes among the

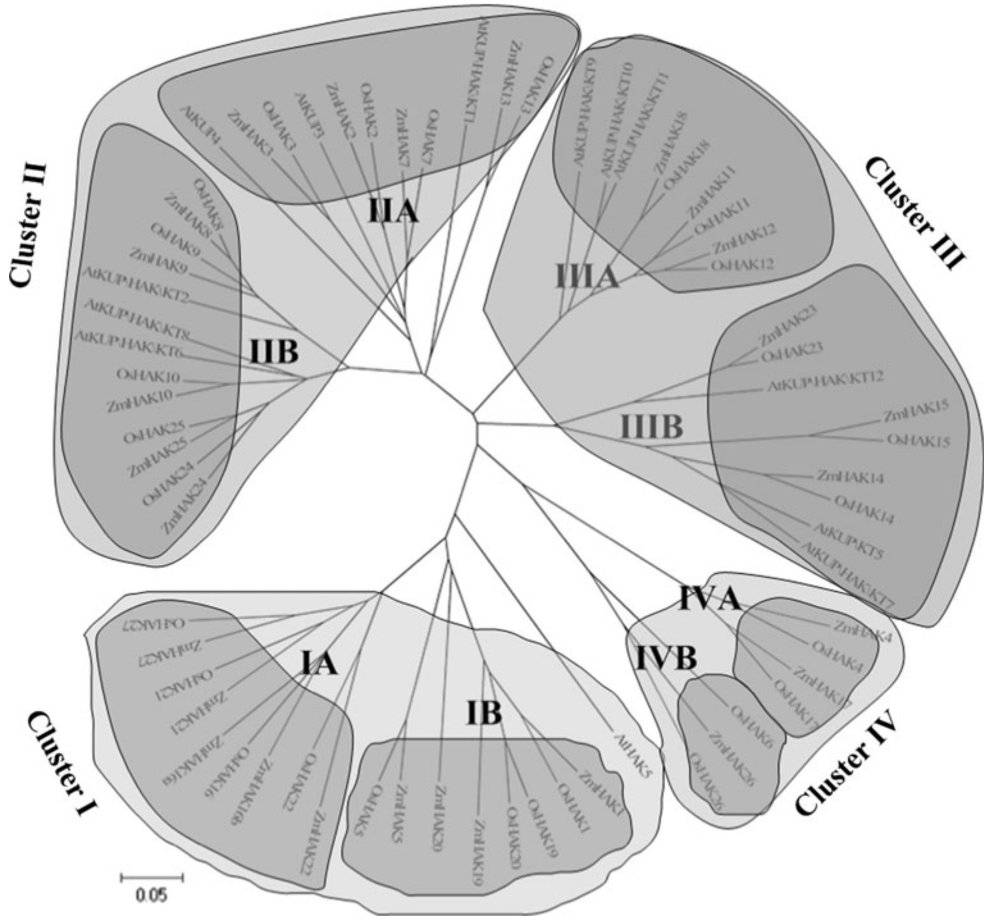

Fig. 4 The phylogenetic tree for *Zea mays*, *Arabidopsis*, *Oryza sativa* HAK potassium transporters. The joint unrooted tree was generated using MEGA4 by the neighbor-joining method. Bootstrap values from 1,000 replicates were indicated at each branch

flanking regions for these pairs of paralogous genes, except for the pair of *ZmHAK8*/*ZmHAK9* (Fig. 1).

Cis-Element analysis

By searching the PLACE database, promoter regions, 2kp range genomic DNA sequences upstream of translation start site of *ZmHAK* genes were analyzed. Amount to 83 putative *cis*-elements more than 6 bp length were identified (Supplementary Table 1). In this study a series of *cis*-elements were found that involved in Ca^{2+}-responsive, abiotic stress responsive, light and circadian rhythms regulation and seed development, such as: ABRERATCAL, ABRELATERD1, ACGTATERD1, ANAERO1CONSENSUS, BOXLCOREDCPAL, CARGCW8GAT, CIACADIANLELHC, CTRMCAMV35S, DPBFCOREDCDC3, EECCRCAH1, E2FCONSENSUS, INRNTPSADB, MYBPLANT, POLASIG2, PRECONSCRHSP70A, RYREPEATGMGY2, RYREPEATLEGUMINBOX, SEBFCONSSTPR10A, SEF4MOTIFGM7S, TATABOX2.

Expression pattern of *ZmHAKs* family

EST data could provide invaluable information for gene expression investigation. The number of maize ESTs takes third place, about 2,019,114 in the NCBI EST database. Thus, EST mining and WGS database were adopted to analyze transcript levels of *ZmHAK* genes. All *ZmHAK* genes had expression data support except *ZmHAK20* and *ZmARF21* hit only one EST. Other 25 *ZmHAK* genes were supported by at least two ESTs. ESTs number of *ZmHAK8* was highest, up to 57. Expression data of three *ZmHAK* genes, including *ZmHAK7*, *ZmHAK16b* and *ZmHAK27*, were checked in only one tissue (Table 2).

Discussion

HAK potassium transporters family

The members of HAK transporter family were firstly found in bacteria [14]. But this family gene did not exist in all organisms. They were present in Plantae, Fungi, probably Amoebozoa and bacteria, but not in Protista and Animalia [29, 45–48]. In Plantae, the homologues of HAK genes existed as multigene families in both monocots and dicots [49]. There were at least 13 members in the *Arabidopsis*, 21 members in poplar, 27 members in rice, and 5 members in barley [11, 12, 34]. In this study, 27 genes belonging to

Table 2 EST expression analysis of *ZmHAK* genes in different tissues

Gene	Tissue and organ type (NCBI)								Number of ESTs in dbEST
	Embryo	Meristem	Seedling	Root	Tassel	Ear	Seed	Multiple	
ZmHAK1		+	+	+				+	20
ZmHAK2	+				+	+	+	+	19
ZmHAK3		+	+	+				+	3
ZmHAK4								+	2
ZmHAK5	+		+	+	+	+	+	+	49
ZmHAK7			+					+	4
ZmHAK8	+		+	+	+	+		+	57
ZmHAK9			+	+	+	+		+	36
ZmHAK10	+		+	+		+		+	14
ZmHAK11		+	+					+	8
ZmHAK12		+	+	+				+	8
ZmHAK13					+	+		+	12
ZmHAK14	+		+					+	15
ZmHAK15	+		+		+	+		+	24
ZmHAK16a		+	+					+	3
ZmHAK16b			+					+	2
ZmHAK17					+	+		+	20
ZmHAK18			+	+	+	+	+	+	17
ZmHAK19								+	2
ZmHAK20									0
ZmHAK21								+	1
ZmHAK22		+	+	+	+	+		+	6
ZmHAK23	+		+	+		+	+	+	18
ZmHAK24								+	2
ZmHAK25	+		+	+	+	+		+	25
ZmHAK26					+	+		+	4
ZmHAK27			+					+	8

the HAK potassium transporter family in maize were identified, as much as that in rice.

Characterization of putative *cis*-elements in the promoter of *ZmHAK* gene family

*C*is-elements played pivotal function in the regulation of gene expression by controlling the efficiency of the promoters. Studies on *cis*-elements could provide key foundation for further functional research of the *ZmHAK* gene family [12]. In this study a series of putative *cis*-elements were found that involved in Ca^{2+}-responsive, abiotic stress responsive, light and circadian rhythms regulation and seed development. Among these putative *cis*-elements, the water-stress responsive element ACGTATERD1 was distributed in all the 27 *ZmHAK* genes [50]. There were other five types abiotic stress responsive *cis*-elements in the promoter of most *ZmHAK* genes, i.e., ABRELATERD1, PRECONSCRHSP70A, SEBFCONSSTPR10A for dehydration responsive [51], ANAERO1CONSENSUS for anaerobic responsive and BOXLCOREDCPAL for wound responsive [52]. Another widely distributed *cis*-element ABRERATCAL, Ca^{2+}-responsive element, was present in 23 of the 27.

In plants, Ca^{2+}, as a secondary messenger, was responded to several environmental stresses and could activate or deactivate a gene by regulating its *cis*-elements [12]. In *Arabidopsis*, some K^+ channel openers (KCO) K^+ channel family genes were activated by cytosolic Ca^{2+} [12]. Similar to ABRERATCAL, four protein storage related elements (DPBFCOREDCDC3, RYREPEATGMGY2, RYREPEATLEGUMINBOX, and SEF4MOTIFGM7S) [53], one light-regulated element, INRNTPSADB [54] and one circadian-regulated element, CIACADIANLELHC [55], also existed in most of the *ZmHAK* gene promoters.

Expression behavior of *ZmHAK* gene family

The members of HAK transporter family in rice, *Arabidopsis*, barley, and common ice plant showed tissue-specific expression patterns. The rice *OsHAK1* was expressed in the whole plant, but was up-regulated in conditions of K^+ deficiency in roots. *OsHAK7* and *OsHAK10* were abundant in seedlings of roots and shoots [24]. In *Arabidopsis* four AtKUP-transporters *AtKUP1-4* were found in roots, stems, leaves, and flowers [18]. The barley *HvHAK1* was only detected in roots [4]. In common ice plant, both *McHAK1* and *McHAK4* were detected in root, leaf, stem, flower, and seed. But *McHAK4* was the most abundant transcript while *McHAK1* was present at low-abundance. *McHAK3*, showed moderate abundance in roots whereas *McHAK2* in the stems [27]. In this study, EST expression analysis of *ZmHAK* genes indicated that *ZmHAK* transporters were expressed in seedling, root, tassel, ear, seed, embryo and meristem. These results suggested that this family did not simply mediated primary K^+ uptake from soil and some transporters might also involve in regulatory and developmental processes.

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